INCH-POUND

A–A–59590/10A
12 August 2005
SUPERSEDING
A-A-59590/10
1 March 2002

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE, WITH VISUAL LOCK INDICATION AND SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/10	B, C, H	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Options

Example: AA59590/10BN38 is the PIN for a black anodize finished, 3.8 inch (96.5 mm) long card holder with visual lock indication and a retained actuating screw. The card holder contains no mounting holes for fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, 5, in table 1, and A–A–59590.

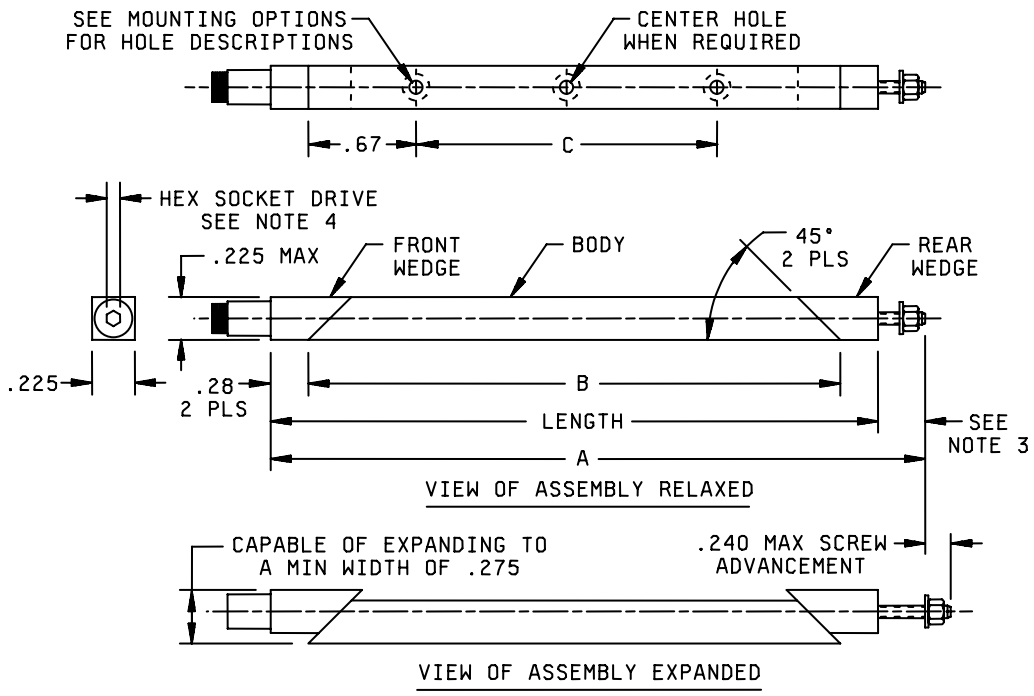
Visual lock indicator (see figure 2). When the card holder is in the unlocked (relaxed) state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

Screw retention (locknut, see figure 3). The screw retention feature intended to prevent unintentional disassembly of the card holder shall be as specified in figure 3. The locknut shall be fabricated of a corrosion resistant material.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.07	1.8	.225	5.72	.28	7.1
.220	5.59	.275	6.99	.67	17.0

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.5 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders without option “W” or “D” and .30 (7.6 mm) maximum on card holders with option “W” or “D”.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

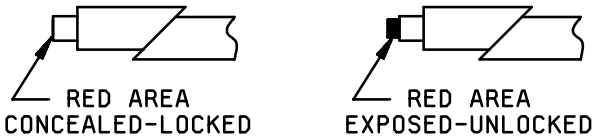


FIGURE 2. Visual lock indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.



FIGURE 3. Screw retaining self-locking nut (corrosion resistant) details.

Finish. The finish options available are as follows: “B” (black anodize), “C” (chemical film), or “H” (hard black anodize) shall be as specified in A–A–59590.

Mounting. The mounting options shall be as specified in A–A–59590. The options available are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). See table I and figure 1 for mounting hole spacing requirements. Card holders using mounting option “J” or “K” may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to 4.45 mm) deep.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5)	Dimension “A” (maximum)	Dimension “B” ± .03 (0.8)	Dimension “C” ± .005 (0.13)
18	1.8 (45.7)	2.20 (55.9)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.70 (68.6)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.20 (81.3)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.40 (86.4)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.70 (94.0)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.90 (99.1)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.20 (106.7)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.70 (119.4)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.20 (132.1)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.70 (144.8)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.20 (157.5)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.70 (170.2)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.20 (182.9)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.70 (195.6)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.20 (208.3)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.70 (221.0)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.20 (233.7)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths are listed in table I.

Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

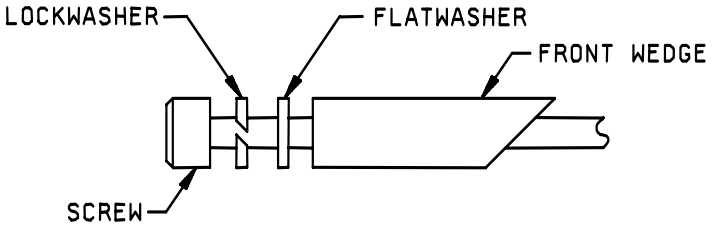


FIGURE 4. Lockwasher and flat washer details.

Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

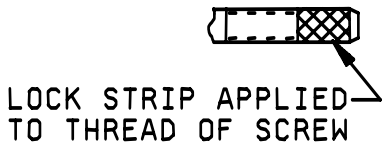


FIGURE 5. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/10	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/10	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	42VI–6–LN	CJ28	V230CR2.80HR
CN28	42VI–6–LN–A	CN28	V230CR2.80R
CT28	42VI–6–LN–T	CT28	V230CR2.80T0R
CV28	42VI–6–LN–S	CV28	V230CR2.80T2R
CG28	42VI–6–LN–M	CG28	MV230CR2.80TM2.5R
CJ28E	42VI–6–LN–L	CJ28E	V230CR2.80HLR
CN28E	42VI–6–LN–A–L	CN28E	V230CR2.80LR
CT28E	42VI–6–LN–T–L	CT28E	V230CR2.80T0LR
CV28E	42VI–6–LN–S–L	CV28E	V230CR2.80T2LR
CG28E	42VI–6–LN–M–L	CG28E	MV230CR2.80TM2.5LR
CJ28W	42VI–6–LF–LN	CJ28W	WV230CR2.80HR
CN28W	42VI–6–LF–LN–A	CN28W	WV230CR2.80R
CT28W	42VI–6–LF–LN–T	CT28W	WV230CR2.80T0R
CV28W	42VI–6–LF–LN–S	CV28W	WV230CR2.80T2R
CG28W	42VI–6–LF–LN–M	CG28W	MWV230CR2.80TM2.5R

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	42VI–6–LF–LN–L	CJ28D	WV230CR2.80HLR
CN28D	42VI–6–LF–LN–AL	CN28D	WV230CR2.80LR
CT28D	42VI–6–LF–LN–TL	CT28D	WV230CR2.80T0LR
CV28D	42VI–6–LF–LN–SL	CV28D	WV230CR2.80T2LR
CG28D	42VI–6–LF–LN–ML	CG28D	MWV230CR2.80TM2.5LR
BJ28	42VI–6–B–LN	BJ28	VA230CR2.80HR
BN28	42VI–6–B–LN–A	BN28	VA230CR2.80R
BT28	42VI–6–B–LN–T	BT28	VA230CR2.80T0R
BV28	42VI–6–B–LN–S	BV28	VA230CR2.80T2R
BG28	42VI–6–B–LN–M	BG28	MVA230CR2.80TM2.5R
BJ28E	42VI–6–B–LN–L	BJ28E	VA230CR2.80HLR
BN28E	42VI–6–B–LN–A–L	BN28E	VA230CR2.80LR
BT28E	42VI–6–B–LN–T–L	BT28E	VA230CR2.80T0LR
BV28E	42VI–6–B–LN–S–L	BV28E	VA230CR2.80T2LR
BG28E	42VI–6–B–LN–M–L	BG28E	MVA230CR2.80TM2.5LR
BJ28W	42VI–6–B–LF–LN	BJ28W	WVA230CR2.80HR
BN28W	42VI–6–B–LF–LNA	BN28W	WVA230CR2.80R
BT28W	42VI–6–B–LF–LNT	BT28W	WVA230CR2.80T0R
BV28W	42VI–6–B–LF–LNS	BV28W	WVA230CR2.80T2R
BG28W	42VI–6–B–LF–LNM	BG28W	MWVA230CR2.80TM2.5R
BJ28D	42VI–6–B–LF–LNL	BJ28D	WVA230CR2.80HLR
BN28D	42VI–6–B–LFLNAL	BN28D	WVA230CR2.80LR
BT28D	42VI–6–B–LFLNTL	BT28D	WVA230CR2.80T0LR
BV28D	42VI–6–B–LFLNSL	BV28D	WVA230CR2.80T2LR
BG28D	42VI–6–B–LFLNML	BG28D	MWVA230CR2.80TM2.5LR
HJ28	42VI–6–B3–LN	HJ28	VHA230CR2.80HR
HN28	42VI–6–B3–LN–A	HN28	VHA230CR2.80R
HT28	42VI–6–B3–LN–T	HT28	VHA230CR2.80T0R
HV28	42VI–6–B3–LN–S	HV28	VHA230CR2.80T2R
HG28	42VI–6–B3–LN–M	HG28	MVHA230CR2.80TM2.5R

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/10	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/10	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	42VI–6–B3–LN–L	HJ28E	VHA230CR2.80HLR
HN28E	42VI–6–B3–LN–AL	HN28E	VHA230CR2.80LR
HT28E	42VI–6–B3–LN–TL	HT28E	VHA230CR2.80T0LR
HV28E	42VI–6–B3–LN–SL	HV28E	VHA230CR2.80T2LR
HG28E	42VI–6–B3–LN–ML	HG28E	MVHA230CR2.80TM2.5LR
HJ28W	42VI–6–B3–LN–LF	HJ28W	WVHA230CR2.80HR
HN28W	42VI–6–B3–LFLNA	HN28W	WVHA230CR2.80R
HT28W	42VI–6–B3–LFLNT	HT28W	WVHA230CR2.80T0R
HV28W	42VI–6–B3–LFLNS	HV28W	WVHA230CR2.80T2R
HG28W	42VI–6–B3–LFLNM	HG28W	MWVHA230CR2.80TM2.5R
HJ28D	42VI–6–B3–LFLNL	HJ28D	WVHA230CR2.80HLR
HN28D	42VI–6–B3LFLNAL	HN28D	WVHA230CR2.80LR
HT28D	42VI–6–B3LFLNTL	HT28D	WVHA230CR2.80T0LR
HV28D	42VI–6–B3LFLNSL	HV28D	WVHA230CR2.80T2LR
HG28D	42VI–6–B3LFLNML	HG28D	MWVHA230CR2.80TM2.5LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/10	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/10	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BN18	42VI–4–B–LN–A	BN18	VA230CR1.80R
BN23	42VI–5–B–LN–A	BN23	VA230CR2.30R
BN28	42VI–6–B–LN–A	BN28	VA230CR2.80R
BN30	3/	BN30	VA230CR3.00R
BN33	42VI–7–B–LN–A	BN33	VA230CR3.30R
BN35	3/	BN35	VA230CR3.50R
BN38	42VI–8–B–LN–A	BN38	VA230CR3.80R

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.

TABLE III. Example of PIN with available length designators – Continued.

PIN designator AA59590/10	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/10	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BN43	42VI–9–B–LN–A	BN43	VA230CR4.30R
BN48	42VI–10–B–LN–A	BN48	VA230CR4.80R
BN53	42VI–11–B–LN–A	BN53	VA230CR5.30R
BN58	42VI–12–B–LN–A	BN58	VA230CR5.80R
BN63	42VI–13–B–LN–A	BN63	VA230CR6.30R
BN68	42VI–14–B–LN–A	BN68	VA230CR6.80R
BN73	42VI–15–B–LN–A	BN73	VA230CR7.30R
BN78	42VI–16–B–LN–A	BN78	VA230CR7.80R
BN83	42VI–17–B–LN–A	BN83	VA230CR8.30R
BN88	42VI–18–B–LN–A	BN88	VA230CR8.80R

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–009

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:20Z
Check the source to verify that this is the current version before use.